Project 02
AI Music & Audio Analyzer

Test & Development Report — v0.8

Item	Current Value
Project Status	In Progress — hardware, DSP, real music analysis and visualization verified
Platform	Windows Laptop
Python	3.14.7
Environment	VS Code + PowerShell + Python venv
Hardware	Windows Laptop + Airdopes Joy Bluetooth Earbuds
Music Input	test_music.mp3
Current Milestone	Step 16 — Loudness, Dynamic Range & Clipping Analysis

1. Project Objective

Build a practical Python-based AI Music & Audio Analyzer using a Windows laptop and Airdopes Joy Bluetooth earbuds. The project progressively validates Bluetooth audio capture, digital signal processing, spectral analysis, frequency-band analysis, rhythm analysis, musical key estimation, loudness analysis and professional visualization.

2. Completed Steps

Step	Feature	Status
1	Audio Device Detection	PASS
2	Airdopes Microphone Verification	PASS
3	Airdopes WAV Recording	PASS
4	WAV Audio Analysis	PASS
5	Waveform Visualization	PASS
6	Audio Level Analysis	PASS
7	FFT / Frequency Spectrum	PASS
8	STFT / Spectrogram	PASS
9	Spectral Feature Analysis	PASS
10	Bass / Mid / Treble Energy	PASS
11	Tempo / Beat Analysis	PASS
12	Integrated Audio Feature Summary	PASS
13	Real Music File Input & Analysis	PASS
14	Professional Music Visualization Dashboard	PASS
15	Key / Chroma Analysis	PASS
16	Loudness / Dynamic Range / Clipping	PASS

3. Airdopes Hardware Validation

Parameter	Result
Device	Airdopes Joy Hands-Free
Input	1 channel / 44,100 Hz
Recording	5.00 sec
RMS Level	-30.15 dBFS
Peak Level	-7.45 dBFS
Crest Factor	13.638

4. Real Music Track Baseline

Parameter	Result
File	test_music.mp3
Duration	240.02 sec
Sample Rate	44,100 Hz
Channels	1 (mono)
Samples	10,585,088
RMS Level	-17.05 dBFS
Peak Level	-0.17 dBFS
Dominant Band	Bass — 89.42%
Estimated BPM	109.96
Detected Beats / Onsets	437 / 1,052

5. Key / Chroma Analysis — Step 15

Feature	Result
Dominant Pitch Class	D
Estimated Key	G Minor
Best Correlation	0.7472
Second Candidate	D Minor
Second Correlation	0.7196
Confidence Gap	0.0276
Reported Confidence	High
Top 3 Candidates	G Minor, D Minor, A# Major
Output	reports\music_key_chroma.png

Interpretation: G Minor is the strongest key candidate using the chroma/profile correlation method, but D Minor is a close alternative because the correlation gap is only 0.0276. Therefore the report treats G Minor as an estimated key rather than ground truth.

6. Loudness / Dynamic Range / Clipping — Step 16

Parameter	Result
RMS Amplitude	0.140450
RMS Level	-17.05 dBFS
Peak Amplitude	0.980979
Peak Level	-0.17 dBFS
Crest Factor	6.985
Peak-to-RMS Dynamic Range	16.88 dB
Average Short-Term RMS	-18.07 dBFS
Maximum Short-Term RMS	-10.10 dBFS
Minimum Short-Term RMS	-102.59 dBFS
Short-Term Level Variation	92.49 dB
Clipping Threshold	0.999
Positive Clipped Samples	0
Negative Clipped Samples	0
Total Clipped Samples	0
Clipping Percentage	0.000000%
Clipping Status	No Clipping Detected
Output	reports\music_loudness_dynamic_range.png

Important interpretation: the 92.49 dB short-term level variation is strongly influenced by very low-level/silent frames and is not a standardized music dynamic-range measurement. The 16.88 dB peak-to-RMS value is retained as a simple signal-level dynamic indicator.

7. Professional Visualization Dashboard — Step 14

The original dashboard was refined into a clearer professional v2 layout with KPI cards, full-track waveform, logarithmic frequency spectrum, frequency-balance bars, focused spectrogram and a dedicated spectral/rhythm information panel.

Output	Purpose
music_analysis_dashboard.png	Original dashboard version retained for project history
music_analysis_dashboard_v2.png	Refined professional dashboard
music_key_chroma.png	Step 15 chroma and key visualization
music_loudness_dynamic_range.png	Step 16 loudness and clipping visualization

8. End-to-End Analyzer Flow

Airdopes / Music File → Audio Loading → Signal Level → Waveform → FFT → Spectrogram → Spectral Features → Bass/Mid/Treble → Tempo/Beat → Integrated Summary → Key/Chroma → Loudness/Dynamic Range → Clipping Detection → Visualization

9. Current Project Structure

Project-02-AI-Music-Audio-Analyzer/

■■■ audio/ → airdopes_test.wav

■■■ music/ → test_music.mp3

■■■ reports/ → analysis graphs, dashboards, key/chroma and loudness reports

■■■ src/ → Steps 1–16 Python analysis scripts

■■■ tests/

■■■ venv/

■■■ requirements.txt

■■■ README.md

10. Overall Validation

Validation	Status
Airdopes Bluetooth microphone accessible	PASS
Airdopes recording	PASS
WAV signal analysis	PASS
Waveform / FFT / spectrogram	PASS
Spectral features	PASS
Bass/Mid/Treble	PASS
Tempo/beat analysis	PASS
Integrated analyzer	PASS
Real music input and analysis	PASS
Professional dashboard	PASS
Key/chroma estimation	PASS

Validation	Status
Loudness/dynamic-range analysis	PASS
Clipping detection	PASS
AI interpretation	PENDING

11. Next Development Step

Step 17 — Audio Quality & Health Score. Combine signal level, clipping, dynamic behavior, spectral balance, rhythm stability and other existing measurements into a transparent overall audio-quality assessment. The score should explain which measurements contributed to the result rather than acting as an unexplained black-box number.

12. Version History

v0.1 — Project-02 documentation baseline created from the beginning.

v0.2 — Steps 4–6: WAV analysis, waveform visualization and audio-level analysis.

v0.3 — Steps 7–8: FFT spectrum and STFT spectrogram.

v0.4 — Step 9: spectral feature analysis.

v0.5 — Steps 10–11: Bass/Mid/Treble and Tempo/Beat analysis.

v0.6 — Steps 12–13: integrated feature summary and real music track analysis.

v0.7 — Step 14: professional music visualization dashboard and refined v2 layout.

v0.8 — Steps 15–16: key/chroma analysis plus loudness, dynamic range and clipping analysis.

Only Project-02 development history is included.